Surya Vamshika Chalapalli

Senior Full Stack & Backend Engineer

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PROFESSIONAL SUMMARY

Senior Full Stack & Backend Engineer with 5+ years of experience designing, developing, and scaling enterprise-grade web applications and SaaS platforms. Strong expertise in Node.js, Express.js, React.js, TypeScript, PostgreSQL, MongoDB, and microservices architecture.

Currently leading the technical execution of a large-scale HRMS platform, driving backend architecture, feature delivery, production readiness, release reviews, and overall engineering quality while mentoring junior developers and interns. Experienced in building payroll systems, authentication platforms, workflow automation, and scalable APIs for high-traffic business applications.

Skilled in modern engineering workflows including CI/CD pipelines, Jenkins-based deployment processes, cloud deployment, production release support, system optimisation, security hardening, and AI-assisted development workflows for faster delivery and improved engineering productivity.

TECHNICAL SKILLS

Languages

- JavaScript (ES6+), TypeScript

Frontend

- React.js, HTML5, CSS3, Tailwind CSS, React Router, TanStack Query, Vue.js

Backend

- Node.js, Express.js, REST APIs, Microservices Architecture

Databases

- PostgreSQL, MongoDB, MySQL, pgvector

Authentication & Security

- JWT, OAuth 2.0, RBAC, SSO, TLS/HTTPS Hardening

DevOps & Cloud

- Docker, Jenkins, CI/CD Pipelines, AWS (EC2, RDS, S3)

Testing & QA

- Jest, Unit Testing, Integration Testing, Postman

Tools

- Git, GitFlow, Bitbucket, VS Code, Jira, Confluence, Claude AI

PROFESSIONAL EXPERIENCE

Senior Full Stack Developer | Incredible Visibility

Jun 2023 – Present

Project: HRMS Platform (Employee Management, Payroll, Attendance, Leave & Document Management)

- Leading the technical execution and feature delivery of a configurable HRMS platform using Node.js, Express.js, TypeScript, React.js, and PostgreSQL.
- Driving backend architecture decisions, API design standards, release planning, production readiness reviews, and engineering quality improvements across the project.
- Acting as the functional technical lead for the project by mentoring 2 junior developers and 1 intern, reviewing implementations, assigning tasks, conducting code reviews, and ensuring delivery quality across modules.
- Built and enhanced core HRMS modules including employee management, payroll processing, attendance tracking, leave management, document workflows, role-based access systems, and approval pipelines.
- Designed and implemented a configurable payroll engine supporting salary structures, statutory deductions (PF, ESI, TDS), reimbursement workflows, and custom pay rules.
- Automated payroll processing, attendance calculations, leave accruals, and scheduled HR workflows using cron-based automation systems, significantly reducing manual HR operations.
- Led improvements to project architecture and overall platform stability to make the application production-ready and scalable for enterprise usage.
- Contributed to the enhancement and scaling of existing microservices-based payroll and attendance systems, improving deployment flexibility, maintainability, and overall platform stability.
- Developed React.js dashboards and reusable frontend modules with optimised API handling using TanStack Query, reducing unnecessary re-fetching and improving application responsiveness.

- Implemented secure authentication and authorisation systems using OAuth 2.0, JWT, RBAC, and multi-role access management for Admin, HR, Manager, and Employee users.
- Added structured API logging, monitoring, validation handling, and error tracking for critical business workflows to improve maintainability and issue detection.
- Participated in CI/CD release workflows using Jenkins for staging and production deployments, including build verification, deployment monitoring, release validation, and issue troubleshooting.
- Worked closely with QA and deployment teams during production releases by analysing logs, debugging deployment issues, validating fixes, and ensuring stable production rollouts across HRMS modules.
- Participated in feature planning discussions, implementation reviews, production issue debugging, and release verification processes.
- Leveraged AI-assisted development workflows for debugging, refactoring, documentation, and accelerating feature development.

Key Achievements

- Improved application scalability and platform stability through backend optimisation, service-level improvements, and enhancements across existing microservices-based systems.
- Reduced REST API response times by approximately 25% through PostgreSQL query optimisation and async processing improvements.
- Successfully delivered multiple enterprise-ready HRMS modules used for payroll automation, attendance management, and employee workflows.
- Improved engineering quality by introducing implementation reviews, structured delivery processes, and development standards across the team.

Backend Developer | Incredible Visibility

Mar 2021 – Jun 2023

Project: Hello Leads (Visitor Analytics, Lead Tracking & Subscription Platform)

- Designed and developed scalable REST APIs using Node.js, Express.js, and MongoDB for a visitor analytics and lead tracking platform processing millions of engagement events.
- Built cron-based data processing pipelines for analytics generation, visitor scoring, reporting aggregation, and scheduled data workflows.
- Integrated Stripe payment gateway for subscription billing, webhook handling, plan management, and payment lifecycle processing.
- Developed internal admin dashboards using Vue.js and Tailwind CSS for platform monitoring, lead funnel analysis, and billing management.
- Optimised MongoDB queries and aggregation pipelines, reducing high-traffic query latency by approximately 30%.
- Collaborated closely with frontend, QA, and product teams to deliver stable production-ready releases.
- Implemented API validation, logging improvements, and backend performance optimisation across analytics services.

Career Break & Skill Development

Jan 2019 – Feb 2021

- Took a planned career break due to family responsibilities while continuing technical upskilling in modern JavaScript and full-stack development technologies.
- Focused on strengthening expertise in Node.js, React.js, REST APIs, authentication systems, databases, and scalable backend architecture.
- Built personal learning projects and improved practical knowledge in modern web application development workflows.

Programmer Analyst | Walmart

Jun 2018 – Dec 2018

Project: Electronic Journal (Enterprise Systems Integration & Cloud Platform)

- Developed Node.js APIs integrating Walmart internal enterprise systems with the OneOps cloud platform for operational automation workflows.
- Replaced legacy authentication systems with OAuth 2.0-based SSO implementation, improving security and user authentication workflows.
- Hardened API security configurations and TLS implementations based on internal security audit requirements.
- Built operational data conversion utilities using Node.js and C++ for enterprise migration workflows.

PROJECTS

Internal Task Tracking Tool

AI-assisted Full Stack Project

- Built an internal task management platform enabling employees and teams to manage assigned tasks, time tracking, status updates, reporting, and workflow visibility.
- Implemented role-based access systems for Developers, QA Engineers, Managers, HR, and Admin users.
- Developed clock-in/clock-out tracking with automatic billable hour calculations and reporting features.
- Added CSV/Excel export functionality and reporting dashboards for management visibility.
- Built using Node.js, Express.js, MongoDB Atlas, React.js, Tailwind CSS, and JWT authentication.
- Leveraged AI-assisted development workflows for rapid prototyping, feature iteration, debugging, and implementation acceleration.

EDUCATION

M.S. Computer & Information Sciences

Kent State University, Ohio, USA | 2016

B.Tech – Computer Science & Engineering

Guru Nanak Institute of Technology, Hyderabad | 2014

ADDITIONAL NOTES

- Open to Senior Full Stack, Backend Engineer, Technical Lead, Product Engineering, and Remote International Opportunities.
- Experience working across architecture planning, implementation reviews, release readiness, production support, and engineering process improvements.